A FIBRED-IN-GROUPOIDS SUBCATEGORY OF A STACK IN GROUPOIDS WITH STACKY LOCAL SECTION PRESHEAF IS A STACK

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $F : \mathcal{S} \to \mathcal{T}$ be a 1-morphism of categories fibred in groupoids over $\mathcal{C}$. Assume that*

(1) *$\mathcal{T}$ is a stack in groupoids over $\mathcal{C}$,*
(2) *for every $U \in \mathrm{Ob}(\mathcal{C})$ the functor $\mathcal{S}_U \to \mathcal{T}_U$ of fibre categories is faithful,*
(3) *for each U and each $y \in \mathrm{Ob}(\mathcal{T}_U)$ the presheaf*

$$(h : V \to U) \longmapsto \{(x, f) \mid x \in \mathrm{Ob}(\mathcal{S}_V), f : F(x) \to f^*y \text{ over } V\}/\cong$$

is a sheaf on $\mathcal{C}/U$.

Then $\mathcal{S}$ is a stack in groupoids over $\mathcal{C}$.

Proof. We have to prove descent for morphisms and descent for objects.

Descent for morphisms. Let $\{U_i \to U\}$ be a covering of $\mathcal{C}$. Let x, x' be objects of $\mathcal{S}$ over U. For each i let $\alpha_i : x|_{U_i} \to x'|_{U_i}$ be a morphism over U_i such that α_i and α_j restrict to the same morphism $x|_{U_i \times_U U_j} \to x'|_{U_i \times_U U_j}$. Because $\mathcal{T}$ is a stack in groupoids, there is a morphism $\beta : F(x) \to F(x')$ over U whose restriction to U_i is $F(\alpha_i)$. Then we can think of $\xi = (x, \beta)$ and $\xi' = (x', \mathrm{id}_{F(x')})$ as sections of the presheaf associated to $y = F(x')$ over U in assumption (3). On the other hand, the restrictions of ξ and ξ' to U_i are $(x|_{U_i}, F(\alpha_i))$ and $(x'|_{U_i}, \mathrm{id}_{F(x'|_{U_i})})$. These are isomorphic to each other by the morphism α_i. Thus ξ and ξ' are isomorphic by assumption (3). This means there is a morphism $\alpha : x \to x'$ over U with $F(\alpha) = \beta$. Since F is faithful on fibre categories we obtain $\alpha|_{U_i} = \alpha_i$.

Descent of objects. Let $\{U_i \to U\}$ be a covering of $\mathcal{C}$. Let (x_i, φ_{ij}) be a descent datum for $\mathcal{S}$ with respect to the given covering. Because $\mathcal{T}$ is a stack in groupoids, there is an object y in $\mathcal{T}_U$ and isomorphisms $\beta_i : F(x_i) \to y|_{U_i}$ such that $F(\varphi_{ij}) = \beta_j|_{U_i \times_U U_j} \circ (\beta_i|_{U_i \times_U U_j})^{-1}$. Then (x_i, β_i) are sections of the presheaf associated to y over U defined in assumption (3). Moreover, φ_{ij} defines an isomorphism from the pair $(x_i, \beta_i)|_{U_i \times_U U_j}$ to the pair $(x_j, \beta_j)|_{U_i \times_U U_j}$. Hence by assumption (3) there exists a pair (x, β) over U whose restriction to U_i is isomorphic to (x_i, β_i). This means there are morphisms $\alpha_i : x_i \to x|_{U_i}$ with $\beta_i = \beta|_{U_i} \circ F(\alpha_i)$. Since F is faithful on fibre categories a calculation shows that $\varphi_{ij} = \alpha_j|_{U_i \times_U U_j} \circ (\alpha_i|_{U_i \times_U U_j})^{-1}$. This finishes the proof. □

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